

Question ID: 58ba7239

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	Hard

Question

The number a is 55% less than the number b . The number b is 320% greater than 160. What is the value of a ?

Correct Answer: 1512/5, 302.4

Rationale

The correct answer is 302.4. It's given that the number b is 320% greater than 160. It follows that b is equal to 160 plus 320% of 160, which can be written as $b = 160 + 3.2(160)$, or $b = 672$. It's also given that the number a is 55% less than the number b . It follows that a is equal to b minus 55% of b , which can be written as $a = b - 0.55b$, or $a = 0.45b$. Substituting 672 for b in this equation yields $a = 0.45(672)$, or $a = 302.4$.